TOXIC TONGUE



Shiv Sena (Shinde) MLA **Sanjay Gaikwad** is never far from controversy. He once announced a bounty for Rahul Gandhi's tongue; on another occasion, he assaulted a canteen worker over the food served to him. Now, amid the severe drought in Maharashtra, he has urged people not to attend the funerals of farmers who commit suicide, claiming such deaths would stop only when their bodies were left out for animals to eat. The remarks have left even his own party leadership gobsmacked.

RANK INJUSTICE





A curious breach of protocol took place in Malda in late October. West Bengal chief secretary **Manoj Agarwal** received a 'guard of honour' during a visit to review Ganga erosion and flooding, while MoS **Joyel Murmu** got no such welcome. This raised eyebrows, as ministers rank above bureaucrats in the official 'warrant of precedence'. Agarwal has been attracting many such courtesies of late. Since taking charge in May, he has been felicitated by rice mill owners and even a garment business house. Clearly, he is a man who loves attention.

SEAT-SHOPPING



There was a time when ex-Bihar CM Nitish Kumar looked towards Uttar Pradesh to expand his political footprint. On November 1, his son **Nishant Kumar** followed that trail, addressing a Janata Dal (United) conference in Varanasi alongside the party's working president, Sanjay Jha. The irony is that the BJP, the JD(U)'s senior ally in Bihar, did not accommodate the party in the 2022 UP election, when it contested 27 seats and won none. So, is Nishant's Kashi visit the opening bid for a seat-sharing deal this time?

Kaushik Deka* with *Prasad Nichenametla, Ashish Misra, Dhaval S. Kulkarni, Arkamoy Datta Majumdar* and *Amitabh Srivastava

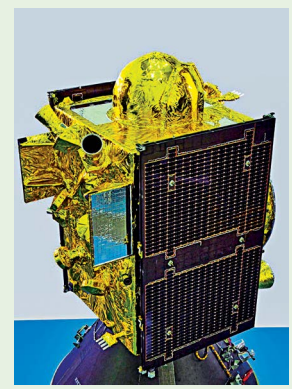
TECH WATCH

INDIA'S NAVIGATION SATELLITES

BACK ON TRACK

By **Ajay Sukumaran**

The successful launch of navigation satellite NVS-03 (*in picture*) from Sriharikota on November 1 achieves the critical objective of restoring India's homegrown satellite navigation constellation called NavIC. This system, providing position, navigation and timing (PNT) services across the country's landmass and up to 1,500 km beyond, had been handicapped after the 10-year-old satellite, IRNSS-1F, reached the end of its mission life in March with its onboard atomic clock ceasing to function. This meant that only three satellites, IRNSS-1B, IRNSS-1I and NVS-01, were available for PNT services though a minimum of four is required for optimal performance. Hence, the NavIC constellation could not provide standalone positioning service though the timing service was functional.



NavIC, earlier known as the Indian Regional Navigation Satellite System (IRNSS), was born out of a critical vulnerability during the Kargil war which prompted India to build an independent navigation satellite system to remove the dependence on foreign systems. It is one of only two regional navigation satellite systems in the world, the other being Japan's Quasi-Zenith Satellite System. NavIC's signals can provide user position accuracy of under 20 metres and timing accuracy of less than 50 nanoseconds, and it is interoperable with global navigation satellite systems. But its ageing fleet and failing foreign-made atomic clocks had become a challenge after two satellites, IRNSS-1H in 2017 and NVS-02 in 2025, could not be placed into orbit.

NVS-03 is the third of a five-satellite series being developed to augment the NavIC constellation. It carries an indigenous atomic clock, along with procured ones. Over the next 10 months, ISRO will launch two more navigation satellites, NVS-04 and NVS-05.